CHAPTER 1

DISCOVERY

DISCOVERY MAY BE the single most exhilarating capacity of the human species. Driven by curiosity, delighted by surprise, we fill the vacuums we perceive and transform the questions we ask into answers. Exploration is so integral to our self-definition that despite manifest dangers and frustrations, we have continued, unrelenting, down its many paths.

Throughout history, human exploration of, in particular, our physical environment has been a story of immense courage in the face of severe risk. Human teams setting out on such endeavors were often in contest with their own mortality. In the early sixteenth century, Ferdinand Magellan's circumnavigation of the globe was a three-year odyssey filled with violence, starvation, and death. Magellan's voyage was the first successfully to steer a course around the globe. In the process, he smashed the endurance record on the high seas, established the dimensions of the planet, and—taking place as the enterprise did in the context of European colonialism—opened the way for social and economic exchange on an international scale.

Most of Magellan's sailors recognized they were risking catastrophe. Though it was no longer widely believed that the Earth was flat, its roundness had not been proved as fact, and many in his crew may have feared sailing off the edge of the world.

Both he and his crew were aware they would pay heavily for any mistaken assumptions. And indeed they did err, and they did pay:

They had underestimated the length of their journey, the quantities of food needed to sustain them, the dangers of mass poisonings, and the risk of damaged or otherwise inoperative ships. Of the original fleet of five vessels carrying approximately 270 crewmembers, only one solitary ship bearing eighteen ghostly survivors returned to port in Spain. Not among them was their captain, who had perished along the way after being hit in the leg by a poisoned arrow.[1]

In the 400 years after Magellan's time, every corner of the world would be mapped—except Antarctica, a land as desolate as an alien planet. The Anglo-Irish explorer Ernest Shackleton would come closer than anyone before him to reaching that land's South Pole: the bottom of the world. In 1909, with an inexperienced crew and with support from no government but only a few private loans and individual contributions, Shackleton and his men set the record for the longest such journey and paved the way for explorers after them.

Although unable to claim he was the first to reach the South Pole itself, Shackleton nevertheless earned the respect of later generations. This he did by prioritizing human values over exploratory ambitions. One year into the expedition, the team members, harnessed daily to sledges for brutal ten-hour shifts, were advancing only a few miles each day. Although they had enough food to reach the Pole, the remaining supply would not suffice to see them back to their ship.

Then and there, within 97 miles of triumph, Shackleton, rather than risking the lives of his men, made the decision to turn back. "We have done our best," he declared in his journal.[2] On their retreat, Shackleton offered his one daily biscuit to Frank Wild, an ailing crew member, who would write in his diary that "All the money that was ever minted would not have bought that biscuit, and the remembrance of that sacrifice will never leave me."[3]

Undeterred by failure, Shackleton would later make further expeditions to the South Pole. For decades, a story (later debunked) circulated that he had placed the following ad in the London *Times*:

> Men Wanted: For hazardous journey. Small wages, bitter cold, long months of complete darkness, constant danger, safe return doubtful. Honour and recognition in case of success.[4]

The ad may have been apocryphal, but the sacrifice it illustrates was not. These were the realities of exploration only a century ago: Our progress on the frontier was limited by the number of brave souls who would accept such long odds.

Perhaps in recognition of these hazards, some governments saw fit to sponsor and reward expeditions of discovery, and these ventures became part of the game of international competition. Magellan's journey, for instance, was defined by politics. Unable to secure financial support from the king of Portugal—his own sovereign—Magellan had defected and sailed instead under the more forthcoming sponsorship of the Spanish crown. After his death, the crew chose a Spaniard, Juan Sebastián Elcano, to assume command. On the return voyage, desperate and virtually bereft of food and supplies, Elcano attempted a stop in the Cape Verde islands—a Portuguese colony off the coast of West Africa—sending thirteen crew ashore to negotiate with the colonial governor. But their request was met with humiliating rejection.

With a broken heart but now even stronger resolve to demonstrate Portugal's folly and Spain's achievement, Elcano ordered the anchor raised and continued onward home. (He left the members of his advance mission behind in unfriendly territory.) Finally completing the journey—and fulfilling Magellan's vision—Elcano wrote to the Spanish king, by now the Holy Roman Emperor and the world's most powerful sovereign:

> Your Majesty will know better than anyone that what we ought most to value and hold on to is that we have discovered and sailed the whole roundness of the world, that going to the West, we have returned from the East.[5]

Eventually, dauntless humans and their political sponsors would push the project of discovery altogether beyond our earthly sphere. We would undertake not just to study the planets—already an ongoing exercise for millennia—but to feel their surfaces beneath our feet. Following the Second World War, the "space race" between the United States and the Soviet Union, a complement of the geopolitical Cold War, brought two superpowers into a competition to send humans where they had never ventured before.

The astronauts themselves could be considered pioneers, but none of the ventures into space undertaken by Moscow and Washington were the product of an individualistic gamble of life and fortune. Rather, each was a coordinated diplomatic and military mission, powered by enormous investments of money, time, and latitude for experimentation. In the past seventy years, more than 600 astronauts have pierced the heavens and toured space, some of whom have gone farther to circle, orbit, or walk on the moon.[6] Thus did U.S.-Soviet rivalry bring us at once to the brink of nuclear annihilation and upward to the stars above.

More than a century before the Age of Discovery—as the early modern era of Magellan, Vasco da Gama, Amerigo Vespucci, and so many others came to be called—

Chinese ambitions on the high seas had been unmatched for their scope and scale. Dwarfing the resources and support provided to explorers by governments in the West, the "treasure fleet" of Zheng He, the great Ming-era admiral, comprised dozens, and sometimes even hundreds, of state-of-the-art ships carrying tens of thousands of sailors, soldiers, diplomats, and merchants. Each of the Chinese admiral's ventures took two years. After navigating the Pacific waters near China's southeastern shores, his ships proceeded westward to the Bay of Bengal, the Indian Ocean, the Arabian Sea, the Red Sea, and finally the Swahili coast of East Africa, in explorations stretching through nearly three decades from 1405 to 1433.[7]

Zheng He's voyages, closer in their origins and motives to the modern American and Soviet space programs than to Western equivalents at his time, were not so much leaps into the unknown as

the rich products of imperial statecraft. But therein also lies a problem. The very things needed to ensure an expedition's success, even within a well-resourced state, can doom its cause in the long run. Politics change. Priorities shift. Human patience is easily exhausted. The expenditures lavished on the "treasure fleet" were so great that factions within the Ming court began to castigate the emperor's funding of these missions. Political setbacks and natural disasters exacerbated the unrest. In the end, the Chinese government decided to destroy or intentionally neglect its best ships, along with many records of Zheng He's voyages, lest a similar visionary arise again to captivate and seduce the country's leadership. The ships rotted away, and the likes of those vessels were not seen again for 400 years.

Analogously, after the U.S. won the space race, without a competitor to motivate its national efforts, support in Washington for space exploration languished and the budget of the National Aeronautics and Space Administration (NASA) was cut. Over the span of five decades, America's capacity for crewed spaceflight steadily degraded. From being the sole nation on Earth capable of landing humans on the moon, the U.S. fell to having the wherewithal only to transport humans to "low Earth orbit," and then, finally, lost the ability to take anyone to orbit at all.

America's reputation was rescued only by the efforts of private explorers who, led by the aerospace company SpaceX, revived cosmic ambitions in the West. That company's efforts have by now led some to the hitherto unimaginable point of viewing our closest celestial neighbors not as temporary destinations, but as permanent homes. A century ago, Shackleton helped establish a human presence on Earth's southernmost pole. Today, Shackleton Crater—a depression, so named in his honor, at the south pole not of Antarctica but of the moon—is the planned site of humankind's next outpost.

The existence of alternative backers has been critical to sustained exploration. In Magellan's Europe, if one monarch proved unsupportive, an explorer could raise financing from another. By the twentieth century, Ernest Shackleton, aspiring to claim the South

Pole for a dwindling British empire distracted by the 1914 outbreak of World War I and unable to raise the necessary funds from the Crown, would turn largely to private donors. The rise of private, for-profit corporations—which allow the pooling of investment and risk—unlocked additional possibilities. One wonders what might have become of Zheng He and his successors if they had had such options.

ENTER AI

For the longest period in Western history, the exploration of reality was focused on geographical entities—our planet and our closest celestial neighbors. As humans steadily acquired mastery over our immediate physical environment—on land, at sea, and in the heavens above—it was only a matter of time before our restless human instinct for discovery would expand its range from the space around us to the ideas within us. Today, we stand at the frontier not of physical but of intellectual exploration.

The development of artificial intelligence has ushered in a new era of discovery. Where AI is integrated into physical systems, robotic sensors assume roles that would earlier have been performed by humans, thereby untethering human discovery from physical danger to the explorer and thus multiplying the ranks of eager developers and investors.

AI also does not experience fear, and so it is undaunted by reality's vast expanse. Nor does it experience shame, and so it unhesitatingly fails—but so quickly can an AI recalibrate that, through constant improvisation and experimentation, it can accommodate high failure rates without causing setbacks to its aforesaid developers and investors.

Today, AI discovery is a project led almost exclusively by private corporations and entrepreneurs, with states emerging as supplementary backers. Even without complementary government action, however, AI growth and expansion are likely to remain fueled

by an abundance of diverse sources of capital. True, in today's still-early phases of development, significant amounts of human capital and societal support may continue to be needed; but sustaining AI exploration in the future may cease to be a fiscal and political drain on the societies that deploy it. Absent unforeseen developments and unlike in earlier ages of exploration—all of which ended before they had achieved their full potential—we can expect discovery of and with AI to continue unabated.

Nevertheless, though AI is partially unshackled from earlier constraints on discovery, it cannot escape them all—particularly as its effects become more pronounced. The tolerance for risk among democratic societies and the uncertain future of international gamesmanship will continue to be a significant X factor in the realm of artificial intelligence. Perhaps the story will be one of an "AI race." Perhaps it will result in the equivalent of the Ming government's wholesale destruction of Zheng He's "treasure fleet." Or, perhaps, states will steer progress along some middle course.

THE POLYMATHIC MIND

In hindsight, it seems obvious that the realm of discovery was bound to expand beyond the physical—the stomping ground of admirals, astronauts, and adventurers—and that its explorers would become more various. And indeed, from fairly early on, history would witness the eventual rise of a new—or, if not new, then decidedly different—kind of human discoverer: the polymath.

Exceptional for their ability to master many spheres of knowledge, any of which might ordinarily absorb a lifetime's labor, polymathic individuals throughout history may number, at a guess, in the mere hundreds. Whether devoted to the arts or to the sciences, or both, all have been imbued with a passion to revolutionize, or to create from scratch, entire fields of study. Propelled not so much by the courage of the heart as by the sheer power of the mind, they have ventured undaunted into the depths of human knowledge and

human imagination: an even vaster terrain than the one confronting a typical explorer of the physical world.

At times, these singular individuals with their wondrous capacities to decode our universe have been viewed with both awe and suspicion as sorcerers, or as putative intermediaries with the divine Creator of the universe—a reputation that could frequently bring them into conflict with religious or political authorities. At other times, treasured for their gifts, they have been encouraged to pursue their studies under the direct auspices of just such authorities, and have found themselves rewarded for their efforts.

In the Islamic Golden Age, polymaths sought ways to pioneer science in service of faith. Ibn al-Haytham, from Basra in modern-day Iraq, proposed the concept of the scientific method—five centuries before Renaissance contemporaries would lay claim to it in the West.[8] Equally at home in geometry, astronomy, optics, and experiential psychology, al-Haytham also had a deep knowledge of hydraulic engineering.

This last would bring him into conflict with religion. Having laid claim to an ability to regulate the flooding of the Nile, a natural phenomenon then still believed to be of supernatural doing, he was invited to meet the caliph in Baghdad. There his proposed engineering projects were found to be at odds with Islamic theology. As punishment for his audacious claims and revolutionary thinking, he was forced into hiding until the caliph's death.

Other polymaths like Muhammad ibn Musa al-Khwarizmi—a Persian from the territory of today's Turkmenistan—found greater success with areas of scholarship in explicit service of their theological masters. Al-Khwarizmi was appointed as court astronomer and head of the library of the House of Wisdom in Baghdad.[9] Astronomy flourished under the Abbasid caliphs, who became lavish patrons of men like him for their direct contributions to the Islamic faith. For example, the geographical coordinates of sacred locations, and most important the direction of Mecca—knowledge of which is a requirement for purposes of Islamic prayer—had been made much more precisely calculable thanks to medieval

astronomy's improved access to the exact positioning of the stars.

Contemporaries of al-Haytham and al-Khwarizmi would find ever more creative ways to sustain the spirit of discovery by deepening the alliance between science and religion, ensuring refuge—for those who possessed, in the words of Ibn Rushd (also known as Averroës), another great Muslim polymath of the same era, "the unity of intellect"—in an age named not for its achievements of reason, as the European Enlightenment was, but for its fervent religiosity.[10]

Thousands of kilometers from Baghdad—and from each other—Chinese and Indian polymaths aligned themselves not with divine authority but with structures of government, thereby sharing a distinctive proximity to politics achieved by both privilege and effort. In the twelfth century, Hemachandra—the "knower of all knowledge in his times"—served as an adviser to King Kumarapala of what is now Gujarat. Centuries later, Akbar the Great—the young Mughal emperor—would both rule the state himself and achieve scholarly successes in architecture, engineering, and literature.

As for Chinese prodigals, in addition to their prowess in intellectual matters, they, too, were deeply engaged in service at court as advisers, scholar-officials, and high-ranking administrators. Government was as much their client as their patron. At any one time of day, these thinkers might be found supervising the formulation of calendrical science in their duties as head officials of the Bureau of Astronomy; an hour later, they could be advising the emperor's central cabinet on how best to increase crop yields. They were commissioned to build grand machines of war, dispatched as ambassadors on diplomatic missions to neighboring kingdoms, and tasked with advising the emperor on matters of economic policy.

But they had no autonomy, being tolerated only to the extent that the emperor sought their intellectual services. In the Chinese world, factional politics became as much a hindrance to genius as clerical restrictions were in the Islamic world. Once-in-a-generation intellects were at the mercy of the very system that had identified their talents in the first place, with science still subservient to the son of heaven. Shen Kuo—a polymath of the Song dynasty—was ousted by a

jealous military officer, ultimately falling out of favor with the emperor and driven into isolation as the outcome of a political rivalry with—strangely enough—the dynasty's only other major polymath.[11]

Lone polymaths flourished during ancient and medieval times in the Middle East, India, and China. But it was not until after the Age of Exploration that systematic conceptual investigation commenced first in Europe and later in the United States, in what we now call the Age of Reason or the Age of Enlightenment. Preceded and facilitated by the fifteenth- and sixteenth-century Renaissance, the ensuing four centuries—bringing us up to the beginnings of our AI age—proved to be a fundamentally different era for intellectual discovery.

In pre-Enlightenment times, polymaths had little choice but to serve a higher power, whether that be an emperor or a caliph. By contrast, many leading figures of the European Enlightenment were enabled to pursue their insights not as means to political or theological ends but as ends in themselves. "A man can do all things if he will," the Italian polymath Leon Battista Alberti proudly claimed of the Renaissance Man.

Still, although intelligence is necessary for exploration, it is not sufficient. Beyond maintaining an appetite for risk, explorers must also be supported by the right resources, the right environment, and the right collaborators. During the Enlightenment, explorers could gain access to all three. Governments and corporations, largely owing to their interest in translating scientific theory into military and commercial applications, remained active patrons and partners to Euro-Atlantic polymaths while, for the most part, freeing the latter to direct their energies and skills as they saw fit. Even when attempts were made to co-opt, suppress, or otherwise intervene, Europe was fragmented enough to allow thinkers of ideas unwelcome in one locale to find a home in another. Thus, the Frenchman François-Marie Arouet, better known by the pen name Voltaire, would spend a significant amount of time outside France, while the Russian Mikhail Lomonosov, having determined at age nineteen to "study sciences," would walk from the constraining circumstances of his native village

in the far north all the way to Moscow, where he acquired a basic education before studying in Kiev and then in Germany at the University of Marburg and then Freiberg.[12]

One salient result was that human progress was now propelled by a newfound concentration and linkage of like-minded thinkers in physical or mental proximity, forcing intelligence of the highest quality into competition and collaboration. Previously, the story of human prodigy had been a lonely one, siloed by the tyrannies of space and time. Often working in isolation, with rarely anyone else in one's spatial or temporal vicinity with whom to collude and commiserate, polymaths could push the boundary only as far as their own capacities would allow. Moreover, this limited connection to the few other intellectual pioneers within and between polities all but ensured a redundancy of effort by inventors lacking direct knowledge of each other's research and unable to build upon each other's work.

Gradually, for those lucky enough to be in possession of accurate, timely, and faithfully translated material, complex inventions could be pieced together by group effort—not just simultaneously but across generations. By the time of the Enlightenment, polymaths were able to bridge not only disciplines but also separate areas of understanding that had never before been reconciled or amalgamated into a coherent whole. No longer was there Persian science or Chinese science; there was just science.

This capability to integrate knowledge from diverse domains helped to produce rapid polymathic breakthroughs, and in time would prove to be the best attempt yet at "collective intelligence." In the twentieth century, for example, the World War II Manhattan Project saw just such a disproportionate density of mental ability as, working together, the era's most brilliant minds translated generations of theoretical physics into devastating application in under five years—a feat that would have been inconceivable to their predecessors. Institutions such as the Institute for Advanced Study at Princeton University and the RAND Corporation in California similarly became sanctuaries for gifted minds.

Naturally, some polymaths, burdened by brilliance, still preferred to work alone. One of them was the Serbian American inventor Nikola Tesla:

> The mind is sharper and keener in seclusion and uninterrupted solitude. No big laboratory is needed in which to think. Originality thrives in seclusion free of outside influences beating upon us to cripple the creative mind. Be alone, that is the secret of invention; be alone, that is when ideas are born. That is why many of the earthly miracles have had their genesis in humble surroundings.[13]

But Tesla was the exception, not the norm. The twentieth century produced a Cambrian explosion of applied science, hurling humanity forward at a speed and scale incomparably beyond previous evolutions. Powered by the combination of their mental capacities, groups of polymaths were also now equipped with the tools of modernity. The aggregate effect of such intense momentum has helped us overcome no small number of human limitations. Digital communication and internet search, themselves the product of polymathic groupings, have enabled the enlargement of such groups and an assembly of knowledge well beyond prior human faculties.

To be sure, there has also been a limit. No matter how well we optimize the design of the vessels intended to carry us to new frontiers, and no matter how well we organize genius into working operations, biological restrictions and human shortcomings continue to constrain our capacities. Our time on Earth is finite. We need sleep. We easily tire. We require rest and respite. Even when on the job, most humans can concentrate on only one task at a time.

Consider, in the late nineteenth and early twentieth century, Thomas Edison—himself a polymath and Tesla's rival—whose thousands of experiments to create a "practical incandescent electric lamp" took three years to achieve success. True, Edison had been partly distracted by contributing improvements to the telephone, the invention of Alexander Graham Bell. But even with the help of many

assistants, his quest for a light bulb required extraordinary effort. Still today, the research and development of many cutting-edge technologies remains expensive, all-consuming, long-drawn-out, and both politically and psychologically daunting to invest in. Because the outcomes are so uncertain, probing the remaining physical frontiers—space, the deep sea, Earth's inner crust—is still the project of only the most successful companies and the wealthiest governments.

Of course, it is precisely these same unfavorable odds that make discovery still feel meaningful, even miraculous.

The physicist John von Neumann, who would later be regarded as among the last great polymaths, was named "Person of the Century" by the *Financial Times* for embodying the twentieth century's characteristic confidence in the power of the mind to "harness and tame the physical world."[14] Indeed, von Neumann had devoted himself with ferocious energy to such central issues as mathematical theory and the atomic bomb, but most crucially to the creation of the computer: both the century's last major breakthrough and among the last inventions that humans ever had to conceive and produce in isolation.

With towering intellects like von Neumann, humanity may have been reaching the upper limits on the ability of non-augmented human intelligence to enlarge our intellectual horizons. Polymathy is particularly rare because ordinarily it takes so long to master the basics of one field that, by the time any would-be polymath does so, he or she will have no time for another and potentially will also have lost the ability to think creatively. Already, innovation today seems increasingly to arise more from teams of people than from any single genius equipped with brilliant cross-disciplinary insight.

Integrating the knowledge gained from multiple individual minds remains, however, a difficult process. Even, or perhaps especially, among star intellectuals, the sheer number of collaborators can hinder synergistic communication.

AI—by contrast—will be the ultimate polymath. In exploring the frontier of human knowledge, it is able to process and generate representations of masses of information at a ferocious rate of

speed. It assesses patterns across countless dimensions and fields simultaneously, creating unprecedented connectivity. Its efficiency allows it to transcend the limitations of human discovery, to the point where it is even expected to succeed in merging many intellectual pursuits into a new "unity of knowledge," in the words of the American sociobiologist E. O. Wilson.[15]

Just as the achievements of Enlightenment polymaths depended on interconnection of information, the recent advances in machine learning have been made possible only by the sheer scale of data—the collective intelligence—that has been not only enabled but made readily accessible by today's AI.

To extend the analogy a step further, perhaps it is no wonder that the latest advancement in AI techniques, rather than relying on a single large program to do the job alone, has synthesized the conclusions of multiple smaller programs in what is known as a "mixture of experts." We predict this will not be the last illustration of the amplified power of polymaths in groups.

The overall project of exploration has, until now, remained constrained by the quantity and quality of humans at the frontier. We have only ever had a few thousand physical pioneers, and a much smaller number of polymaths. Artificial intelligence, thus, is poised to create a revolution in both physical and intellectual exploration. AI, as we have noted, lacks fear and shame, and so unfeelingly runs to the frontier as bidden. Moreover, equally equipped to explore kilometers of outer space and nanometers of human biology, AI's probing of reality is notably unconstrained by subjective experience or physical labor, by human brainpower or human senses. Nor, when it comes to physical reality, does machine exploration ask us to sacrifice our lives to the cause; on the contrary, it might well demand much *less* of our time—that is, only the time measured by what humans provide for it.

In the future, the main constraint on any given society may no longer be the number of talented polymaths it can muster to provide the small and sometimes inconsistent engine of scientific progress. No longer will humanity's potential be capped by the total number of

Magellans or Teslas we produce. Nor may the world's strongest nation any longer be the one with the most Albert Einsteins and J. Robert Oppenheimers, so long as that nation can create and then utilize AI to its fullest potential. And that raises the possibility of a paradigm shift in the primary standard for measuring national strength, which has moved through the centuries from territory, to resources, to capital, to human capital—and now, perhaps, to computing capital.

Moreover, a learning machine could well become a self-improving machine. In the end, then, might the last polymathic invention—namely, computing, which amplified the power of the human mind in a way fundamentally different from any previous machine, and decades later would facilitate breakthroughs in artificial intelligence—be remembered for replacing its own inventors?

A THIRD AGE OF DISCOVERY

From the perspective of AI, the accumulated knowledge of humankind is like an archipelago of volcanic islands spread out atop a boundless ocean. In this imagined scheme, each island's geographical center is dominated by a volcanic peak: As the viewer's gaze slopes downward toward the sea, certainty fades, dropping into lower gradients of confidence until one reaches the seafront.[16]

Assuming that, for purposes of this imaginative exercise, a sufficient quantity of water will have been drained from the Earth's oceans, one would then immediately see a vast topology of underwater terrain hitherto all but invisible to the human eye; no longer do islands appear immediately as free landmasses floating adrift in the ocean, but rather as merely the exposed rocky outcroppings of gigantic underwater mountains or volcanoes—which, rising from their submerged base on the ocean floor, are just tall enough to break the surface.

If, in this scheme, each island is taken to represent a discipline of human understanding, the water that separates each from the next

represents the incomplete connections that still must be uncovered in order to advance our understanding of the universe as a potentially coherent whole. Although we may feel temporarily secure in the mapping of our immediate reality, we have little conception of what lies beneath us, or beyond us. AI could change this.

Take the realm of physics, the quintessential scientific example. If Isaac Newton harmonized the laws of the celestial and the terrestrial worlds, and if Michael Faraday and James Clerk Maxwell did the same for electricity, magnetism, and optics, the search still continues for a "Grand Unified Theory" that will reconcile the two separate and incompatible theories vying to explain our existence from opposite ends of reality. These are the cosmic theory (general relativity) and the subatomic theory (quantum mechanics).

AI might finally bring order and structure to seemingly disparate areas of understanding, exposing in the process (as in archipelagos with the same superstructure) an interconnectedness between such realms as genetics, linguistics, cosmology, and psychology. AI might even help reconcile the divide between and among apparently incompatible schools of thought or systems of belief.

In many disciplines, we have already identified a wide band of possible truths, although many of them have low probabilities of actual truthfulness. In the archipelago of human understanding, these are dots along the shoreline: not ignorance, but not necessarily knowledge. Guided to an area of inquiry along that shore, AI can judge with extraordinary accuracy the most fruitful avenues for further exploration. Selecting, testing, reversing, and reselecting in quick succession, it can assess the effects of millions of possible choices.

That is the method (as we explore further in Chapter 5) by which Google's DeepMind laboratory could not only master the ancient Chinese board game of Go as humans knew it but could also, to the extent that the machine demonstrated its knowledge to humans, add to our own knowledge of the game. As compared with earlier chess-playing programs, which often relied on brute-force computing, AlphaGo, having previously "trained" itself on thirty million prior moves, displayed the ability for machines to reason

abstractly.[17]

In this sense, the machine's training resembled the "training" of the mind of a PhD student of philosophy: in the latter case, a gradual process of building the capacity to think and reason through years of intensive study. Like a student emerging from those years to answer questions at a defense of his or her dissertation, DeepMind's system "trained" itself to transcend the learnings it had previously been exposed to and produce—from its more abstract and higher-level indoctrination—the chess moves that it inferred to hold the highest probability of winning. At times, indeed, the AI model succeeded in selecting moves that had never previously been attempted by a human in 4,000 years of play—possible only because, while the human mind seems to be confined to the manipulation of just four independent variables at a time, AI can bring to bear countless probabilistic judgments from innumerable dimensions at once.[18] AI thus accessed original ideas and, for the first time, brought them inside the bounds of human experience.

A human user querying an AI model—for instance, by typing a question into ChatGPT—is asking it not merely to retrieve a point of information, as conventional search engines do, but to synthesize multiple points of information and on that basis to deliver a conclusion. Moving simultaneously in multiple directions and multiple dimensions, it generates representations of information in high-dimensional space, involving relationships within and among innumerable fields and subfields; and from those complexly networked representations, it derives its conclusions.

Here, in responding to our questions, resides the gift of the seemingly superhuman capacity and speed of AI's "large language models," which are pre-trained on vast amounts of data. As the accuracy of the answers determines the level of our confidence in the various truths we humans espouse, such models produce an increasingly detailed geology of the deep.

AI is likely to accrue new knowledge not only with great speed but also in ways that would leave open a range of additional exploration. In its chess-playing game, AlphaGo tended to gravitate

toward solutions of peculiar openness. Some AI models may thus, in their training, have absorbed a bias toward areas offering a high potential for many options, thereby enabling quick and flexible probing forward.

It might be hard for humans to adjust to this new mode of AI exploration. The most serious challenge will involve whether and how such exploration reflects—or contradicts—our perception of reality and our human purpose. Humans may try to build vehicles to keep up and follow along as we instruct AI to create new outcroppings. Or we may provision ourselves with industrial machinery, by means of which—slowly, at our more human pace—we might dredge sediment from the ocean floor and thereby expand our tiny rock of understanding. Or, perhaps, we might well end up persuaded never again to set foot on land beyond our own.

And then there is the challenge of an improperly controlled AI, which could accumulate knowledge destructively. Its methods of discovery could be as violent as the volcanic events that formed our home in the first place. To stretch the present metaphor: In once more erupting our volcano, the AI could lay waste to prior knowledge while in the process greatly expanding the islands' area. It might even rupture enormous plates along the seafloor, whose collision could bring to the surface new mountains of knowledge—which, however, by virtue of being disconnected from our own experience, might in turn trigger a storm of cognitive crises and thereby lead us ineluctably to a more complete—though unwelcome—understanding of reality.

On the other hand, if aligned with our objectives, then, as previously in the development of computing, the development of AI will be a human mission that can facilitate every other human mission. That would promote AI to the position of the chief or at least the coequal such power in the universe, partially if not wholly responsible for the majority of significant discoveries in the next century. Should that be the case, we humans might come to realize, in retrospect, how small an island we have cultivated over the last few millennia, compared with the peaks of possibility under our feet.